

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1}{}_{\dagger} & \begin{array}{c} \frac{1}{2}(-Y_1 k^2 - 3 \overset{(2)}{K}_3) \\ 0 \end{array} \\
\mathcal{K}_{0^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{c} 0 \\ 0 \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#1}{}_{\alpha} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} \\
\mathcal{K}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} -k^2 \overset{(4)}{K}_2 - \overset{(2)}{K}_3 \\ \frac{Y_2 k^2 + 2 \overset{(2)}{K}_3}{2\sqrt{2}} \end{array} \\
\mathcal{K}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} \frac{Y_2 k^2 + 2 \overset{(2)}{K}_3}{2\sqrt{2}} \\ \frac{1}{2}(-Y_3 k^2 - \overset{(2)}{K}_3) \end{array}
\end{array}
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{c} -k^2 \overset{(4)}{K}_3 \\ 0 \end{array} \\
\mathcal{K}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{c} 0 \\ 0 \end{array}
\end{array}
\end{array}$$

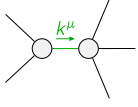
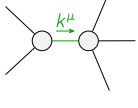
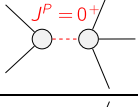
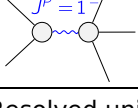
$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1}{}_{\dagger} & \begin{array}{c} \frac{1}{\text{Det}(0^+)} \\ 0 \end{array} \\
\mathcal{J}_{0^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{c} 0 \\ 0 \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#1}{}_{\alpha} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} 0 \\ 0 \end{array} \\
\mathcal{J}_{1^+}^{\#1}{}_{\dagger}{}^{\alpha} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} \frac{-Y_3 k^2 - \overset{(2)}{K}_3}{2 \text{Det}(1^+)} \\ \frac{-Y_2 k^2 - 2 \overset{(2)}{K}_3}{2\sqrt{2} \text{Det}(1^+)} \end{array} \\
\mathcal{J}_{1^+}^{\#2}{}_{\dagger}{}^{\alpha} & \begin{array}{c} 0 \\ 0 \end{array} & \begin{array}{c} \frac{-Y_2 k^2 - 2 \overset{(2)}{K}_3}{2\sqrt{2} \text{Det}(1^+)} \\ \frac{-k^2 \overset{(4)}{K}_2 - \overset{(2)}{K}_3}{\text{Det}(1^+)} \end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta} & \begin{array}{c} -\frac{1}{k^2 \overset{(4)}{K}_3} \\ 0 \end{array} \\
\mathcal{J}_{2^+}^{\#1}{}_{\dagger}{}^{\alpha\beta\chi} & \begin{array}{c} 0 \\ 0 \end{array}
\end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
Y_1 &= 3 \left(\overset{(4)}{K}_1 + \overset{(4)}{K}_2 \right) + 2 \overset{(4)}{K}_3 \text{ \&\& } Y_2 = 2 \overset{(4)}{K}_2 + \overset{(4)}{K}_7 \text{ \&\& } Y_3 = \overset{(4)}{K}_2 + \overset{(4)}{K}_3 + \overset{(4)}{K}_7 \text{ \&\& } \\
\text{Det}(0^+) &= \frac{1}{2} (-k^2 (3 \left(\overset{(4)}{K}_1 + \overset{(4)}{K}_2 \right) + 2 \overset{(4)}{K}_3) - 3 \overset{(2)}{K}_3) \text{ \&\& } \text{Det}(1^+) = \frac{1}{8} (k^4 (4 \overset{(4)}{K}_2 \overset{(4)}{K}_3 - \overset{(4)}{K}_7^2) + 4 k^2 \overset{(4)}{K}_3 \overset{(2)}{K}_3)
\end{aligned}$$

Lagrangian

$$\begin{aligned}
& \overset{(2)}{K}_3 \mathcal{K}_{\alpha}^{\alpha\beta} \mathcal{K}_{\beta\chi}^{\chi} + \overset{(4)}{K}_1 \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} + \overset{(4)}{K}_2 \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} + \\
& \overset{(4)}{K}_3 \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \overset{(4)}{K}_3 \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} + \overset{(4)}{K}_7 \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} = 0$	1	$ \begin{aligned} & \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\beta\chi\delta} + \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\delta\beta\chi} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\alpha\beta\delta} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\delta\alpha\beta} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\beta\alpha\chi} = \\ & \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\alpha\chi\delta} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\delta\alpha\chi} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\beta\alpha\delta} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\alpha\beta\chi} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\chi\alpha\beta} \end{aligned} $	
$\mathcal{J}_{1^+}^{\#2\alpha\beta} = 0$	3	$\partial_{\delta} \partial_{\chi} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + \partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\chi\alpha\beta} = \partial_{\delta} \partial_{\chi} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta}$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} = 0$	3	$\partial_{\delta} \partial_{\chi} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + \partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\beta\alpha\chi} = \partial_{\delta} \partial_{\chi} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta} + \partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} = 0$	5	$ \begin{aligned} & 3 \partial_{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_{\epsilon} \partial^{\epsilon} \partial^{\chi} \partial^{\alpha} \mathcal{J}^{\delta\beta}_{\delta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\beta} \mathcal{J}^{\alpha\chi\delta} + 4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\beta} \mathcal{J}^{\delta\alpha\chi} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\chi} \mathcal{J}^{\beta\alpha\delta} + \\ & 4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\chi} \mathcal{J}^{\delta\alpha\beta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\delta} \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\alpha} \mathcal{J}^{\delta}_{\delta}{}^{\epsilon} + 3 \eta^{\alpha\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial_{\delta} \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\epsilon} \mathcal{J}^{\delta\alpha}_{\delta} = \\ & 3 \partial_{\epsilon} \partial_{\delta} \partial^{\chi} \partial^{\beta} \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_{\epsilon} \partial^{\epsilon} \partial^{\chi} \partial^{\beta} \mathcal{J}^{\delta\alpha}_{\delta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\beta\chi\delta} + 4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\delta\beta\chi} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\chi} \mathcal{J}^{\alpha\beta\delta} + \\ & 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\delta} \mathcal{J}^{\beta\alpha\chi} + 4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\delta} \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\beta} \mathcal{J}^{\delta}_{\delta}{}^{\epsilon} + 3 \eta^{\beta\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial_{\delta} \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\epsilon} \mathcal{J}^{\delta\beta}_{\delta} \end{aligned} $	
Total # constraint(s):	12		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	0	$ \begin{aligned} & \frac{1}{\overset{(4)}{K}_3^2 \overset{(2)}{K}_3} \\ & (8 \overset{(4)}{K}_3^2 p^2 + 9 \overset{(4)}{K}_7^2 p^2 - 3 \overset{(4)}{K}_3 (\overset{(2)}{K}_3 - 6 \overset{(4)}{K}_7 p^2) - \sqrt{(208 \overset{(4)}{K}_3^4 p^4 + 81 \overset{(4)}{K}_7^4 p^4 - 120 \overset{(4)}{K}_3^3 p^2 (\overset{(2)}{K}_3 - 6 \overset{(4)}{K}_7 p^2) + 54} \\ & \overset{(4)}{K}_3 \overset{(4)}{K}_7^2 p^2 (-\overset{(2)}{K}_3 + 6 \overset{(4)}{K}_7 p^2) + 72 \overset{(4)}{K}_3^2 (7 \overset{(2)}{K}_3^2 - 3 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 11 \overset{(4)}{K}_7^2 p^4))}) \end{aligned} $
	1	0	$ \begin{aligned} & \frac{1}{\overset{(4)}{K}_3^2 \overset{(2)}{K}_3} \\ & (8 \overset{(4)}{K}_3^2 p^2 + 9 \overset{(4)}{K}_7^2 p^2 - 3 \overset{(4)}{K}_3 (\overset{(2)}{K}_3 - 6 \overset{(4)}{K}_7 p^2) + \sqrt{(208 \overset{(4)}{K}_3^4 p^4 + 81 \overset{(4)}{K}_7^4 p^4 - 120 \overset{(4)}{K}_3^3 p^2 (\overset{(2)}{K}_3 - 6 \overset{(4)}{K}_7 p^2) + 54} \\ & \overset{(4)}{K}_3 \overset{(4)}{K}_7^2 p^2 (-\overset{(2)}{K}_3 + 6 \overset{(4)}{K}_7 p^2) + 72 \overset{(4)}{K}_3^2 (7 \overset{(2)}{K}_3^2 - 3 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 11 \overset{(4)}{K}_7^2 p^4))}) \end{aligned} $
	1	$-\frac{3 \overset{(2)}{K}_3}{3 (\overset{(4)}{K}_1 + \overset{(4)}{K}_2) + 2 \overset{(4)}{K}_3}$	$-\frac{2}{3 \overset{(4)}{K}_1 + 3 \overset{(4)}{K}_2 + 2 \overset{(4)}{K}_3}$
	3	$\frac{4 \overset{(4)}{K}_3 \overset{(2)}{K}_3}{-4 \overset{(4)}{K}_2 \overset{(4)}{K}_3 + \overset{(4)}{K}_7^2}$	$\frac{4 \overset{(4)}{K}_3^2 + 4 \overset{(4)}{K}_3 \overset{(4)}{K}_7 + 3 \overset{(4)}{K}_7^2}{4 \overset{(4)}{K}_2 \overset{(4)}{K}_3 - \overset{(4)}{K}_3 \overset{(4)}{K}_7}$
Resolved unitarity condition(s):		(Demonstrably impossible)	